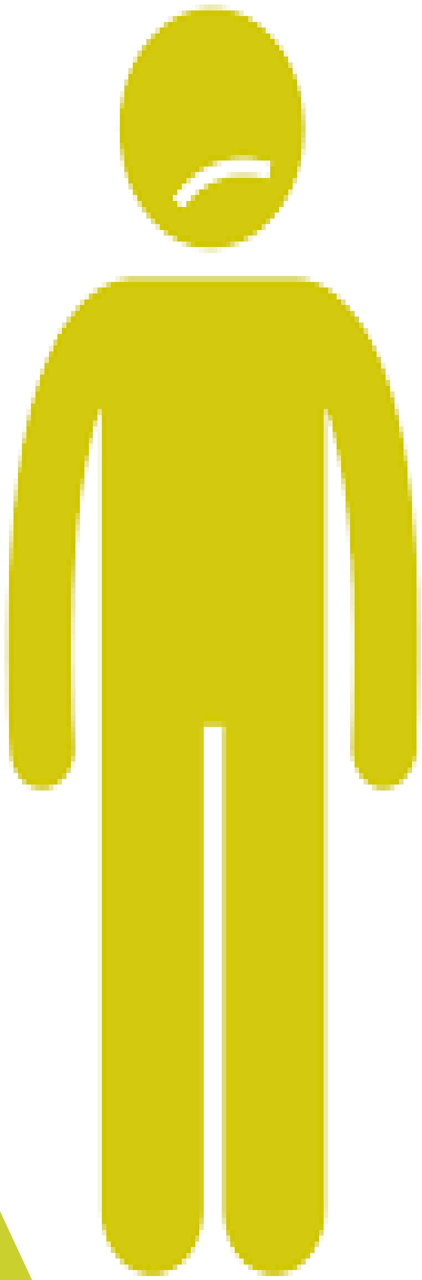
Jaundice and related disorders

Rehab A Omran

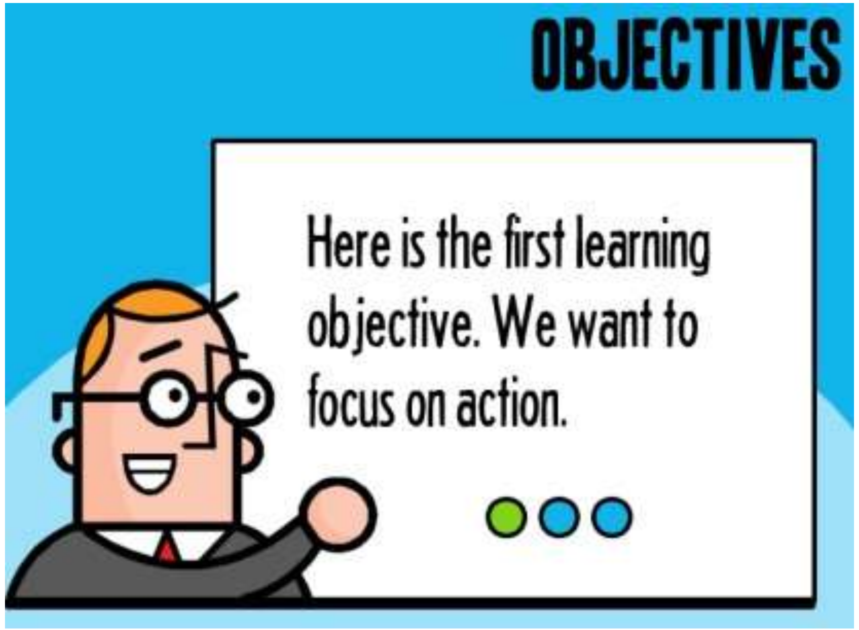


Learning Outcomes

By the end of this presentation you should be able to:

- Define jaundice
- Classify different types of jaundice
- List different causes of jaundice
- Outline clinical approach to a patient with jaundice

Order relevant investigation for a patient with jaundice
Describe different types of viral hepatitis
List high risk groups for hepatitis
Recognize hepatitis B serology
Describe prevention of viral hepatitis
Recognize dental aspects of hepatitis



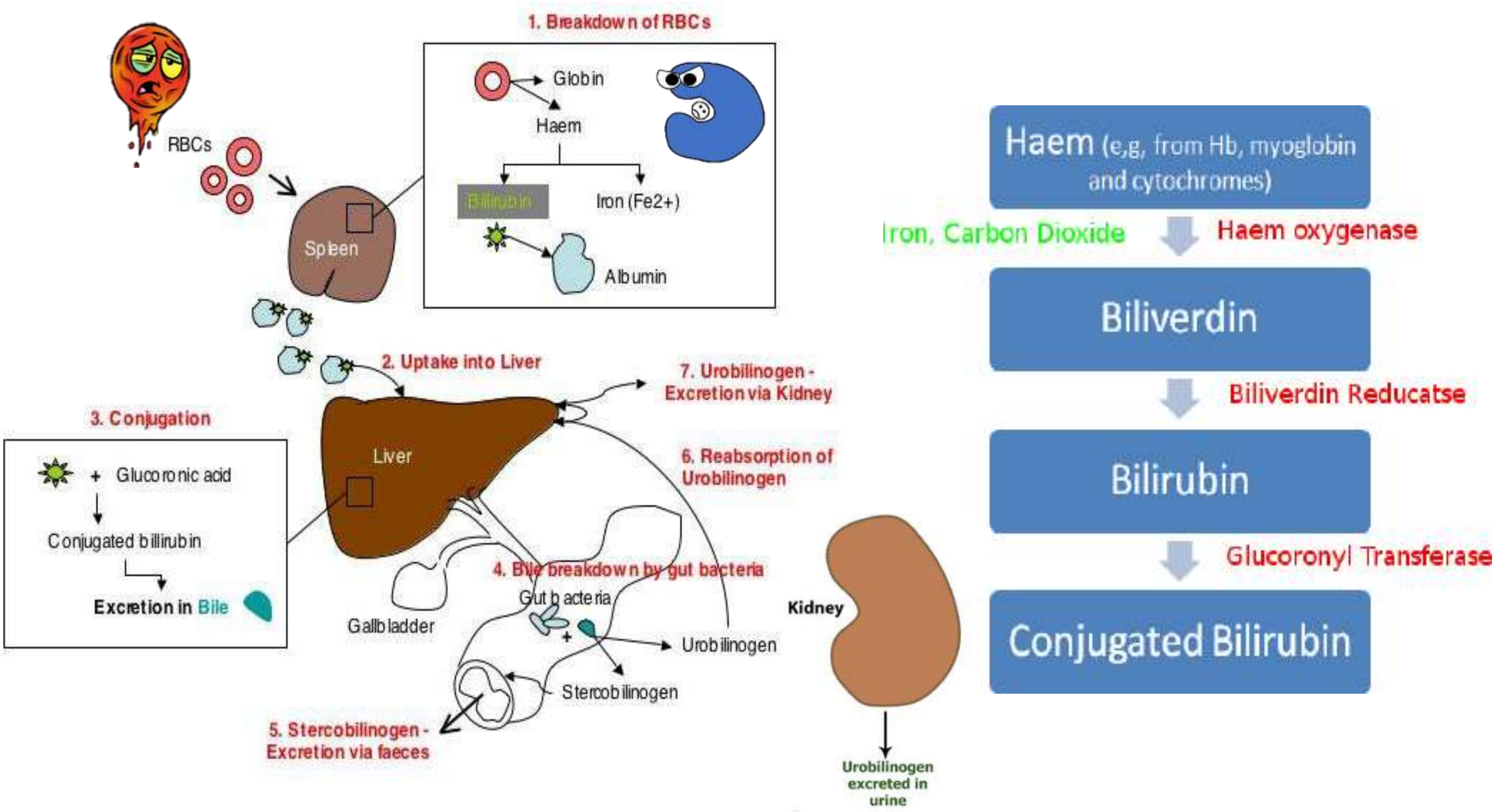
Yellowish discoloration of sclera

Jaundice

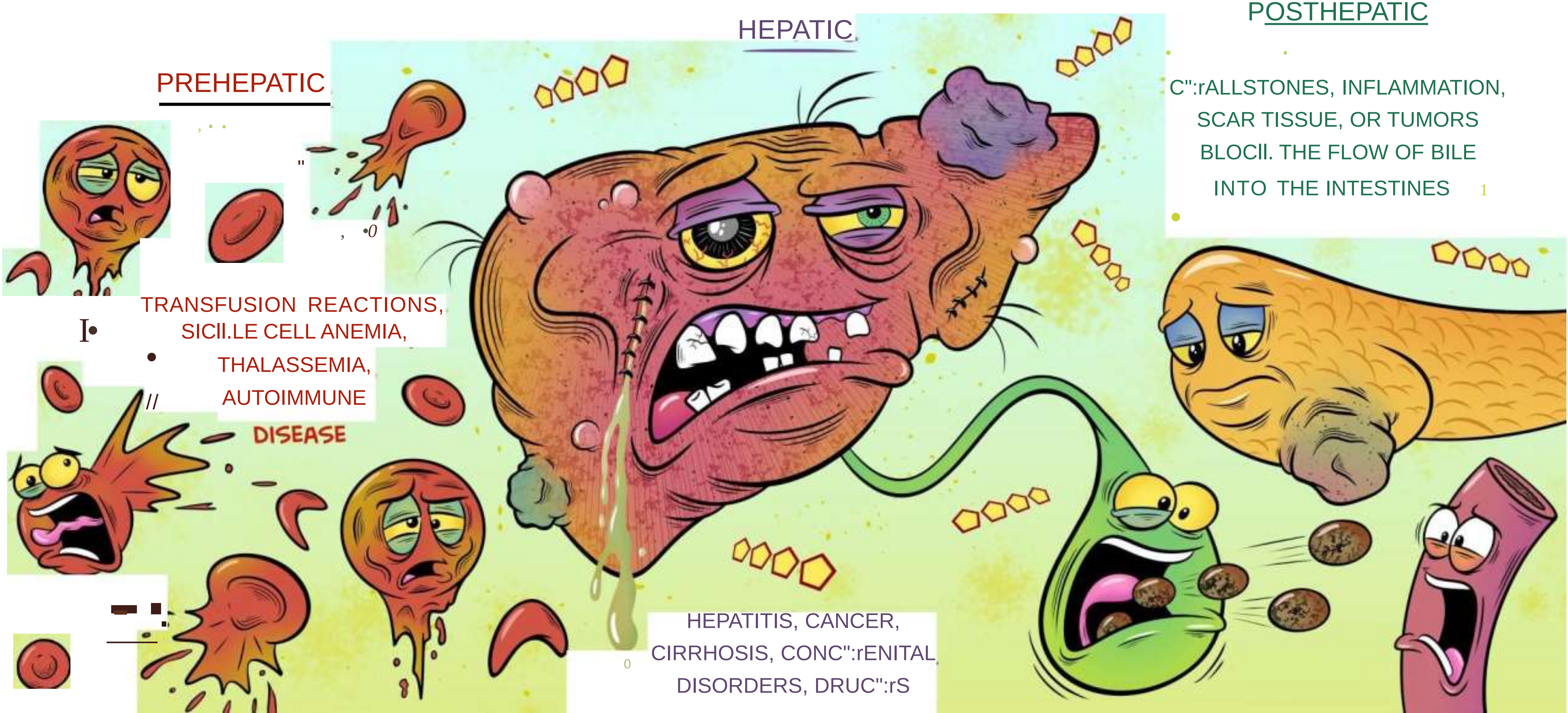
Definition

- Jaundice is defined as a yellowing of skin, mucous membranes and sclera due to the deposition of yellow- orange bile pigment i.e. bilirubin.
- The discoloration typically is detected clinically once the serum bilirubin level rises above **2.5 mg/dL** (40 µmol/L)

Bilirubin Metabolism


1. Breakdown of RBCs
RBCs
Globin
Haem
Bilirubin
Iron (Fe2+)
Albumin
Spleen
2. Uptake into Liver
Liver
3. Conjugation
+ Glucoronic acid
Conjugated billirubin
Excretion in Bile
7. Urobilinogen - Excretion via Kidney
6. Reabsorption of Urobilinogen
4. Bile breakdown by gut bacteria
Gut bacteria
Gallbladder
Urobilinogen
Stercobilinogen
5. Stercobilinogen - Excretion via faeces
Kidney
Urobilinogen excreted in urine
Haem (e,g, from Hb, myoglobin and cytochromes)
Iron, Carbon Dioxide
Haem oxygenase
Biliverdin
Biliverdin Reducatse
Bilirubin
Glucoronyl Transferase
Conjugated Bilirubin

TYPES OF JAUNDICE



PREHEPATIC

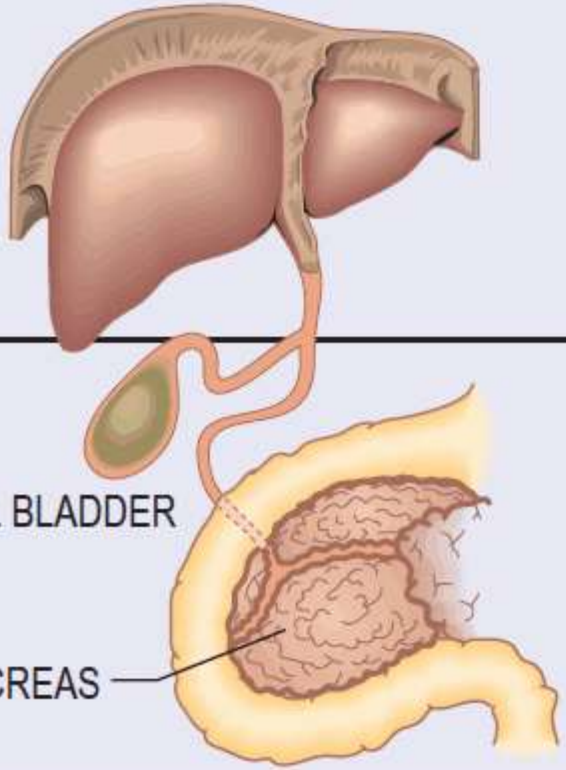
TRANSFUSION REACTIONS,
SICll.LE CELL ANEMIA,
THALASSEMIA,
AUTOIMMUNE
DISEASE

HEPATIC

HEPATITIS, CANCER,
CIRRHOSIS, CONC":rENITAL
DISORDERS, DRUC":rS

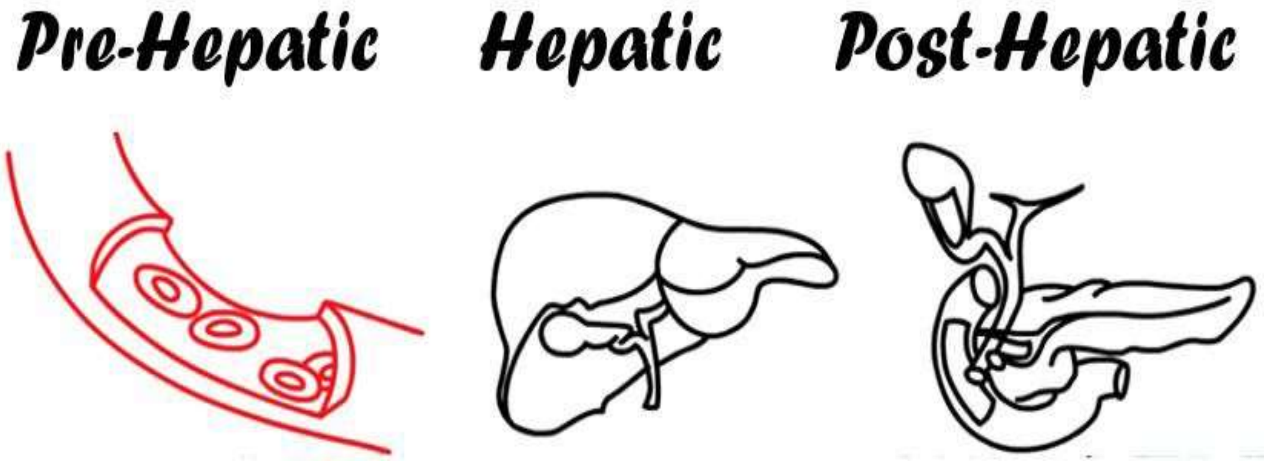
POSTHEPATIC

C":rALLSTONES, INFLAMMATION,
SCAR TISSUE, OR TUMORS
BLOCll. THE FLOW OF BILE
INTO THE INTESTINES

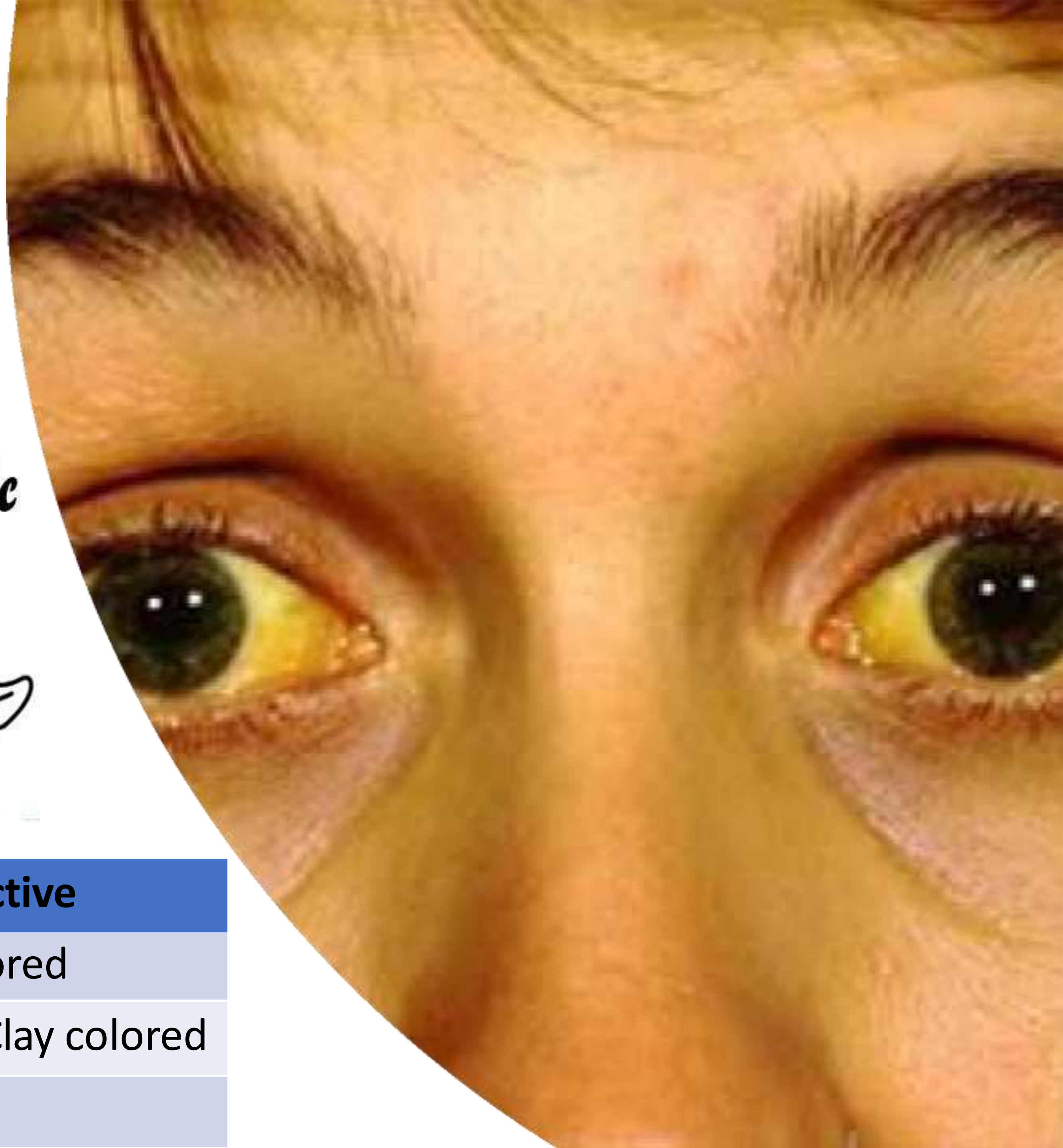
Types	HAEMOGLOBIN	Causes
Prehepatic	BILIRUBIN	Haemolysis
Cholestatic	CONJUGATION	
Intrahepatic		Viral hepatitis Drugs Alcoholic hepatitis Cirrhosis – any type Pregnancy Recurrent idiopathic cholestasis Some congenital disorders Infiltrations
Extrahepatic	GALL BLADDER PANCREAS	Common duct stones Carcinoma – bile duct – head of pancreas – ampulla Biliary stricture Sclerosing cholangitis Pancreatitic pseudocyst

Causes of jaundice.

Types of Jaundice



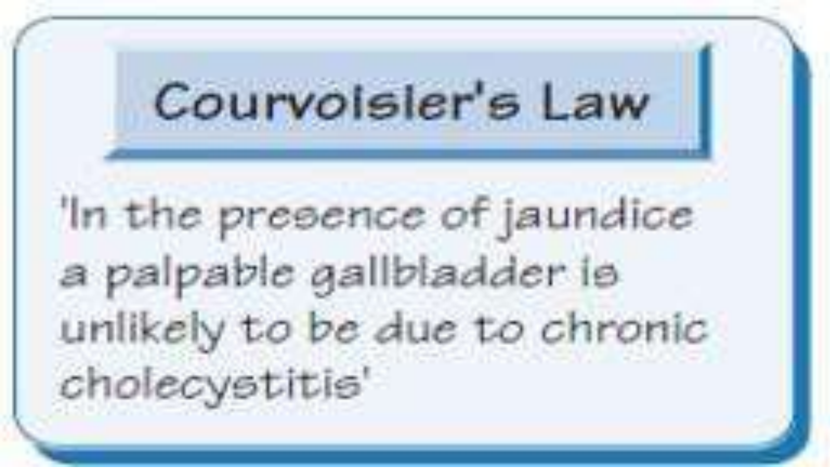
	Pre-hepatic	Hepatocellular	Obstructive
Urine	Normal	Tea colored	Tea colored
Stools	Normal	Normal - Pale	Pale – Clay colored
Itching	No	Mild	Severe



Clinical Approach

Physical Examination

- **Pallor, splenomegaly (**Hemolytic anemia**)**
- **Stigmata of chronic liver disease:**
  - Ascites, LL edema
  - Leukonychia (Hypoalbuminemia)
  - Bruises (Low clotting factors)
  - Palmer erythema, spider nevi, gynecomastia, loss of axillary & pubic hair, testicular atrophy (high estrogen level)
  - Asterixis, poor mental status
- Right hypochondrium tenderness, Murphy’s sign, Charcot triad
- Hepatosplenomegaly, distended gallbladder



Clubbing

DUPUYTREN'S CONTRACTURE

Palmar erythema

Asterixis

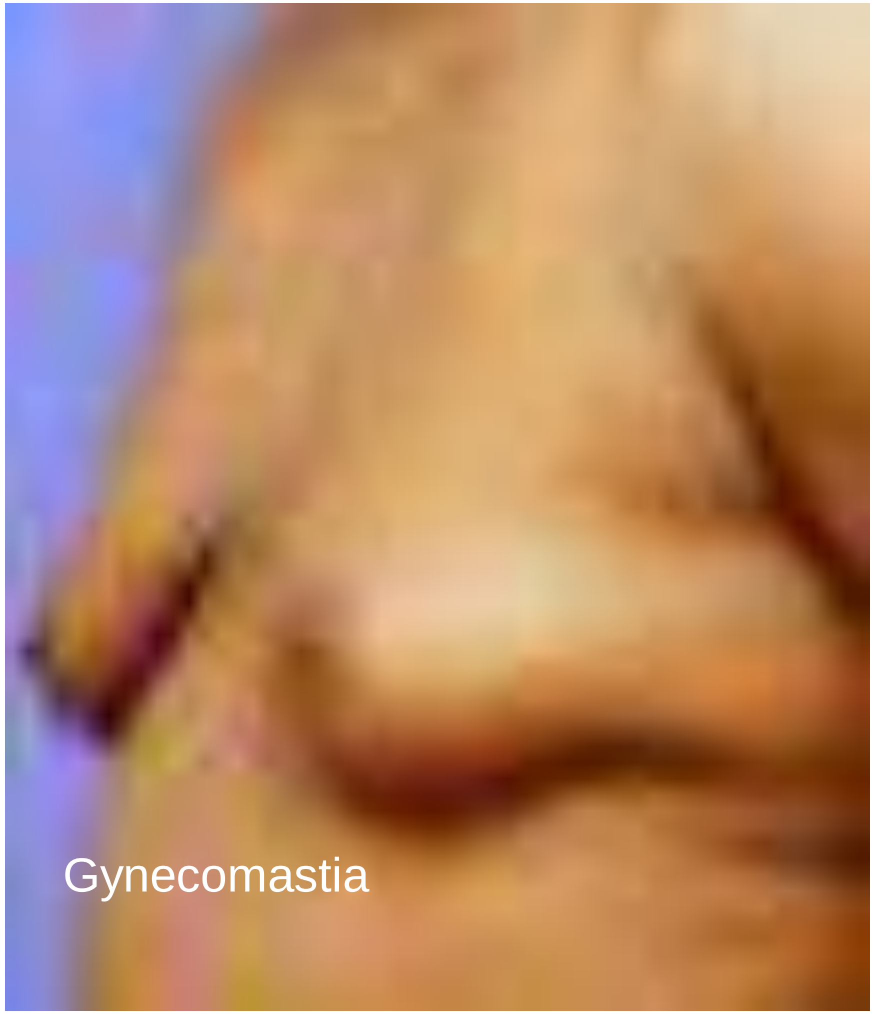
Leukonychia


PALMAR ERYTHEMA

Arms

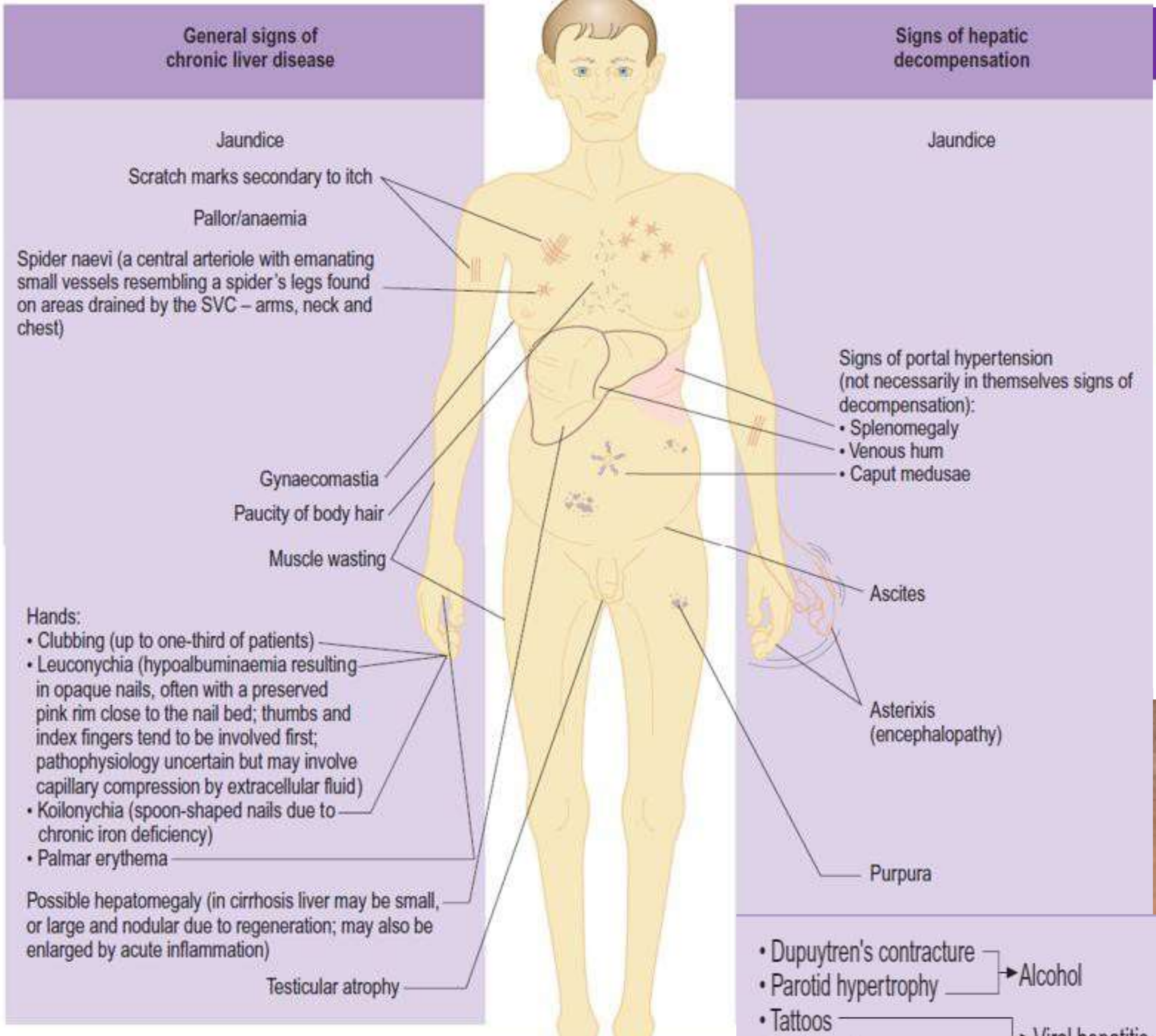
- Scratch marks: obstructive jaundice (particularly primary biliary cirrhosis).
- Bruising: hepatic impairment.
- Needle marks (Antecubital fossa)
- Tattoos


Spider telangiectasias (naevi)


Gynecomastia

Legs

- Pitting edema
- Erythema nodosum
- Pyoderma gangrenosum



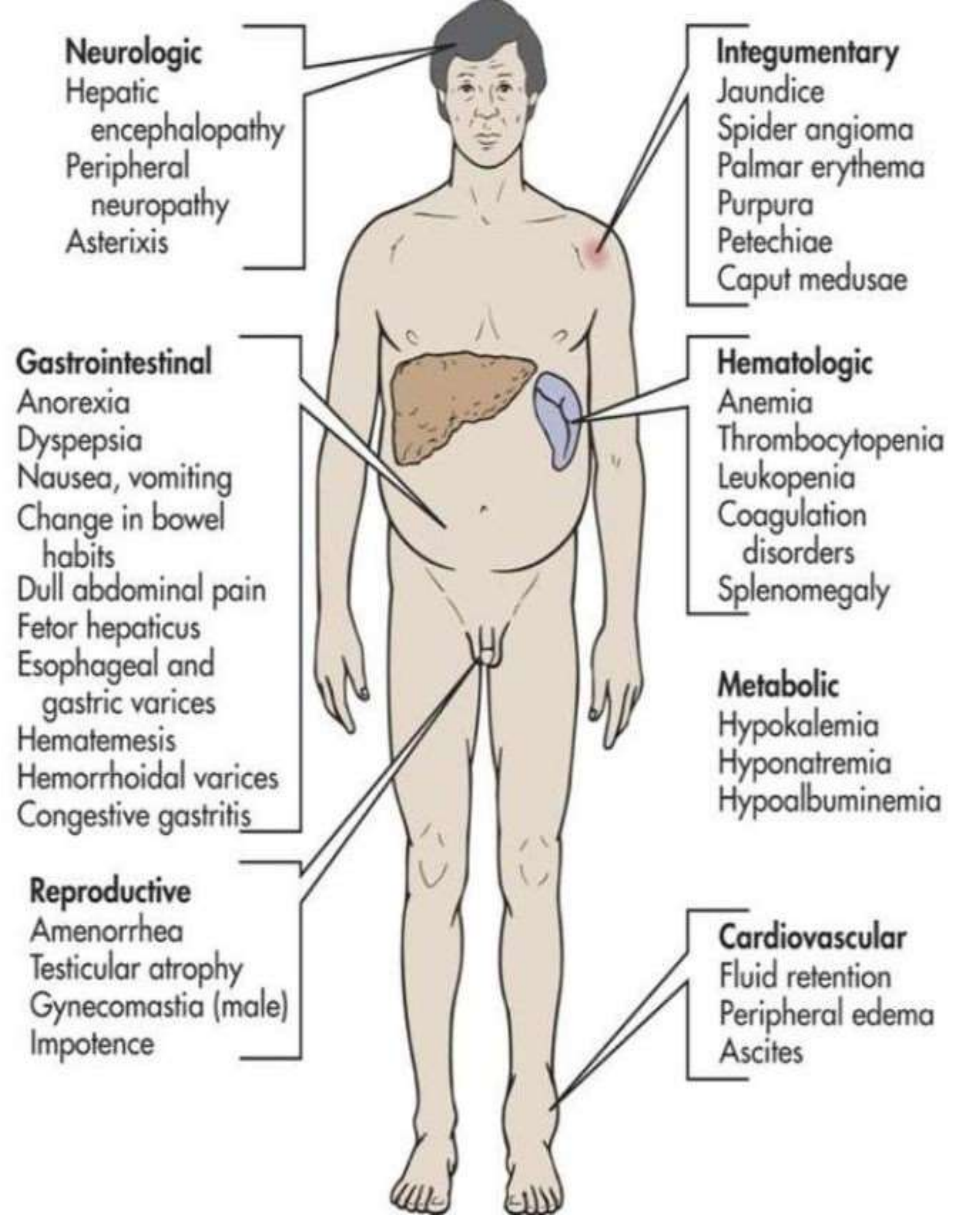
Stigmata of Chronic Liver Disease

Palmar erythema | Spider nevi | Parotid enlarged | Gynecomastia

Muscle atrophy | Asterixis | Dupuytren's contracture | LL edema

Signs which suggest an underlying cause:

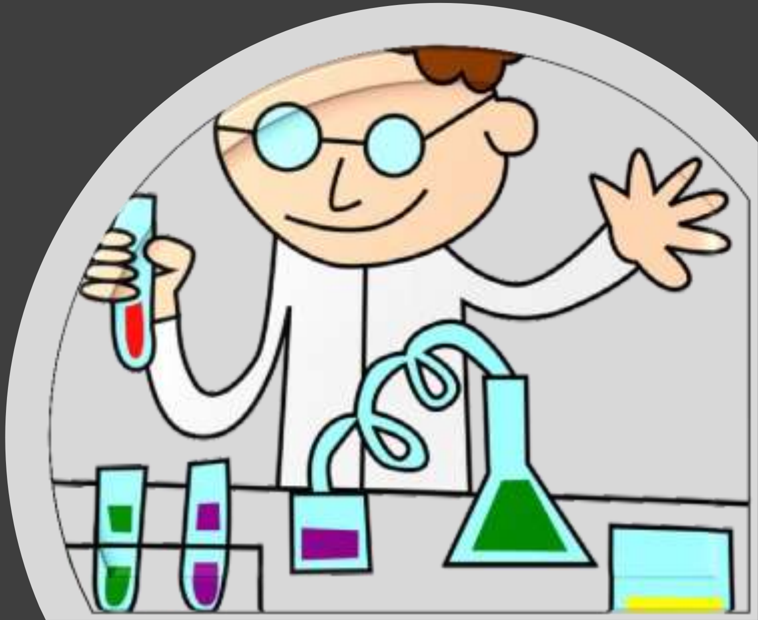
- Dupuytren's contracture, Parotid hypertrophy → Alcohol
- Tattoos, Track marks → Viral hepatitis
- Signs of associated autoimmune disease → Autoimmune hepatitis
- Xanthelasmata → Primary biliary cirrhosis
- Slate grey pigmentation → Haemochromatosis
- Kayser–Fleischer rings → Wilson's disease
- Signs of obstructive airways disease → α_1-antitrypsin deficiency


Neurologic
Hepatic encephalopathy
Peripheral neuropathy
Asterixis
Gastrointestinal
Anorexia
Dyspepsia
Nausea, vomiting
Change in bowel habits
Dull abdominal pain
Fetor hepaticus
Esophageal and gastric varices
Hematemesis
Hemorrhoidal varices
Congestive gastritis
Reproductive
Amenorrhea
Testicular atrophy
Gynecomastia (male)
Impotence
Integumentary
Jaundice
Spider angioma
Palmar erythema
Purpura
Petechiae
Caput medusae
Hematologic
Anemia
Thrombocytopenia
Leukopenia
Coagulation disorders
Splenomegaly
Metabolic
Hypokalemia
Hyponatremia
Hypoalbuminemia
Cardiovascular
Fluid retention
Peripheral edema
Ascites

Investigations

INVESTIGATIONS

- **Basic**
  - **Labs:**
    - **CBC, peripheral smear, reticulocyte counts, LDH, haptoglobin** electrolytes, urea, Cr, glucose.
    - **Bilirubin (conjugated and unconjugated) AST, ALT, ALP, INR, albumin.**
    - HAV IgM, HAV IgG, HBsAg, HBsAb, HBcIgM, anti-HCV,
    - ANA, anti-smooth muscle antibody (ASMA), anti-mitochondrial antibody (AMA).
    - ferritin, ceruloplasmin, a1-antitrypsin, AFP.
  - **Imaging—U/S, CT abdomen**
- **SPECIAL**
  - Endoscopic U/S
  - MRCP
  - ERCP
  - Liver Biopsy



CHARACTERISTICS OF VIRAL HEPATITIS

	HAV	HBV	HCV	HDV	HEV
Genome	RNA	DNA	RNA	RNA	RNA
Family	Picorna	Hepadna	Flavi	viroid	
IP (days)	15-45	30-180	15-150	30-80	15-60
transmission	Feco-oral	Blood Saliva	Blood Saliva	Blood	Feco-oral
Serologic diagnosis	Ig M anti-HAV	Ig M anti-HBV	Anti-HCV HCV RNA	Ig M anti-HDV	Ig M anti-HEV
Carrier state	No	Yes	?	Yes	No
Chronicity	-	+	+	+	-
Prevention	Vaccine	vaccine		Vaccination against HBV	

Hepatitis B

- Hepatitis B, a major cause of liver damage and liver cancer, is a silent epidemic worldwide. **It is a blood borne infection that is commonly transmitted through blood sexual fluids and vertically.**

- Periodontal disease, severity of bleeding, and bad oral hygiene were associated with the risk of HBV.

- **In the dental office**, hepatitis B is mostly **transmitted via blood**. It is vital to ensure infection control practices are being followed during dental procedures such as cleanings, extractions, injections, root canals, and surgical procedures to reduce the incidence of hepatitis B.

- unvaccinated dental **health care workers have a 10 times greater risk** of becoming infected with hepatitis B because of possible occupational exposure

SEROLOGICAL PATTERN OF HEPATITIS B

COMMON SEROLOGIC PATTERNS IN HEPATITIS B VIRUS INFECTION AND THEIR INTERPRETATION

HBsAg	Anti- HBs	Anti- HBc	HBeAg	Anti - HBe	Interpretation
+	-	IgM	+	-	Acute hepatitis B
+	-	IgG	+	-	Chronic hepatitis B with active viral replication
+	-	IgG	-	+	Chronic hepatitis B with low replication
-	-	IgM	+ or -	-	Acute hepatitis B
-	+	IgG	-	+ or -	Recovery from hepatitis B (immunity)
-	+	-	-	-	Vaccination (immunity)
-	-	IgG	-	-	False positive or infection in remote past

Hepatitis C

- Less widespread, rarely transmitted during dentistry.
- **Less readily transmitted by needle stick injuries.**
- More vulnerable to antiseptics.
- **Acute hepatitis is uncommon, usually mild. But:**
- **High risk of HCC (10 %).**
- **Infection persists in 80%.**
- **No protective vaccine.**
- More common in IV drug addicts, tattooing.
- Most HCV patients had higher HCV RNA levels in their gingival sulcus than in their saliva.
- HCV-RNA could be found in tooth brushes used by hepatitis C patients

Diagnosis based on elevated ALT/AST and positive serum **anti-HCV**

Confirmed by detectable HCV-RNA in serum

Oral manifestations of hepatitis B and C infection

lichen planus

Sjögrens syndrome

Sialadenitis,

Some forms of oral cancers may also be seen.

Cirrhotic patients may have thrombocytopenia due to hypersplenism or treatment with interferon.

In patients with liver disease, the resultant impaired hemostasis can be manifested as petechiaes or excessive gingival bleeding with minor trauma. **This is especially suggestive if it occurs in the absence of inflammation**.

lichen planus

Sialadenitis

How to prevent

- All health care providers, including those working in dentistry, be **vaccinated** to protect them. **Patients** can also play a role by ensuring that they are **vaccinated as well**. The hepatitis B vaccine is safe and effective and protects for a lifetime!

- failure to heat-sterilize hand pieces between patients, a lack of training for interns/personnel, and the use of a combination of unsafe injection practices. Help transmission

- The transmission of hepatitis B in dental surgery can be prevented by the routine exercise of **good clinical hygiene**.
- The dental clinic must **properly dispose** of needles, sterilize instruments, and comply with all standard precautions (e.g., **wearing appropriate personal protective equipment** and **disinfecting all equipment and surfaces after each patient) for all patients.**

- Understanding of the basic principles of infection control is mandatory

PRECAUTIONS

- **If a patient with active hepatitis, positive-HBsAg (HBV carrier) status, or positive HCV status requires emergency treatment, use the following precautions**

- Consult the patient's physician regarding status
- If bleeding is likely during or after treatment, measure prothrombin time (PT) and bleeding time.
- All personnel in clinical contact with the patient should use full barrier technique, including masks, gloves, glasses or eye shields, and disposable gowns

- Aseptic techniques should be followed at all times. Minimize aerosol production by not using ultrasonic instrumentation, air syringe, or high-speed hand pieces.
- Remember that saliva contains a distillate of the virus. Pre-rinsing with chlorhexidine gluconate for 30 s is highly recommended

PRECAUTIONS

- Use as many disposable covers as possible, covering light handles, drawer handles, and bracket trays. Headrest covers should also be used
- All disposable items (e.g., gauze, floss, saliva ejectors, masks, gowns, gloves) should be placed in **a lined wastebasket**.
- After treatment, these items and all disposable covers should be bagged, **labeled,** and disposed of, following proper guidelines for bio-hazardous waste
- When the procedure is complete, all equipments should be scrubbed and sterilized. If an item cannot be sterilized or disposed of, it should not be used
- All working surfaces and environmental surfaces should be wiped with 2% activated glutaraldehyde **(Cidex).**

Post-Exposure Prophylaxis

Step one: Treatment of the site of exposure

The site of exposure to infectious fluids should be washed as soon as possible using soap and water only,

- The exposed mucus membranes should be washed with water only.
- Eyes should be flushed with water and saline solution (if there was contact with potentially infectious fluids).
- Do not use caustics and do not rinse the wound with antiseptics and disinfectants.

Step two: Report and documentation

- Occupational exposure should be reported immediately. Details regarding the circumstances under which the exposure happened and the employee was administered prophylaxis should be recorded in the employee's medical record.
- The documentation should include: Date and time of exposure, details of the accident (where and how the exposure happened, the site of exposure on the body ,type and amount of fluid or material a person was exposed to), severity of injury, and details about the source of infectious material. it is necessary to record the details of the exposed person (HBV vaccination, other medical conditions and medications used

Post-Exposure Prophylaxis

Step three: Evaluation of sources

- Test the patient for anti-HBsAg, HCV and HIV antibodies
- Evaluation of "viral load" (the level of infectious particles in the blood)

Test the patient by rapid HIV test

Step four : Specific prophylaxis

- Any blood or body fluid exposure to an unvaccinated person **should lead to the initiation of the hepatitis B vaccine series** (at 0, 1, and 6 months). When **hepatitis B Immune Globulin** (HBIG) is indicated, it should be administered **as soon as possible** after the exposure (preferable **within 24 h**,
- Hepatitis B vaccine can be administered simultaneously with HBIG but at a separate site.
- **During the testing period**, Exposed person should abstain from changes in sexual activity, pregnancy, breastfeeding, or professional activities not to donate blood, plasma,
- **Testing for anti-HBs antibodies 1-2 months after the last dose of vaccine**

Dental Complications of Sickle Cell Disease

- Dental Caries
- Dental Erosions
- Infractions
- Hypodontia
- Malocclusions
- Pulp Necrosis
- Abnormal Trabecular spacing
- Infection

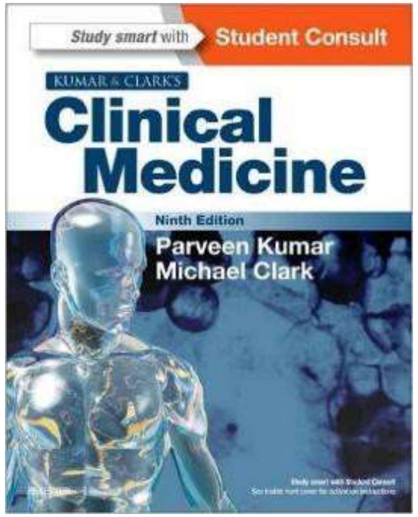
Figure 1: Class I malocclusion in a patient with SS.

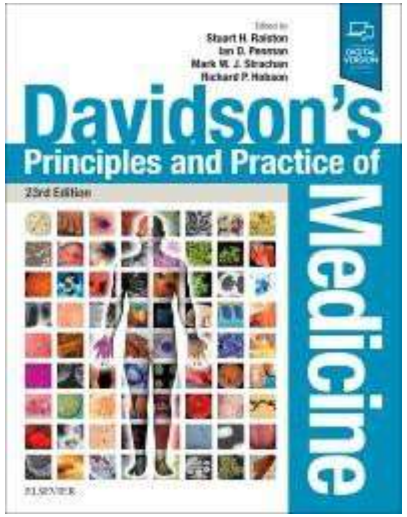
Dental Considerations and Management of Sickle Cell Patients

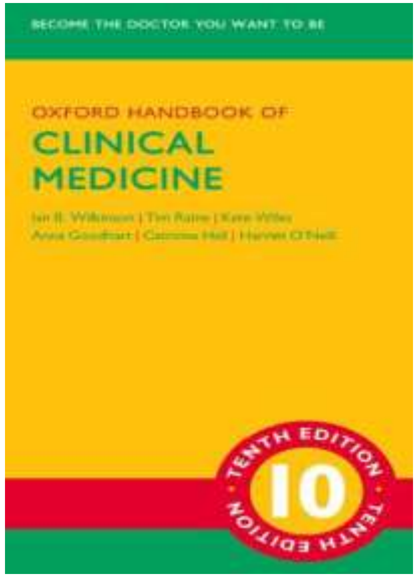
- For sickle cell patients, the most preferred safe analgesic treatment is a combination of paracetamol and codeine. In addition, morphine and its derivative analgesics can be used Aspirin should not be used to avoid the risk of acidosis.
- Extreme care should be taken regarding infections. Any source of infection either dental or non-dental source should be treated vigorously with SCD patients to decrease the chance of crisis caused by infections.
- **prophylactic antibiotics should be prescribed for them for prevention of infection**, if they will undergo special dental operation
- The pulp therapy is a case selection and it is not always a good option especially in the case of the possibility of treatment failure. At that point, extraction is the best option to eradicate the chances of failure. The latter treatment option should be exerted with extreme care to preserve as well as protect the fragile, osteoporotic bone from fracture. In addition, to decrease the risk of future osteonecrosis that may accompany traumatic extraction

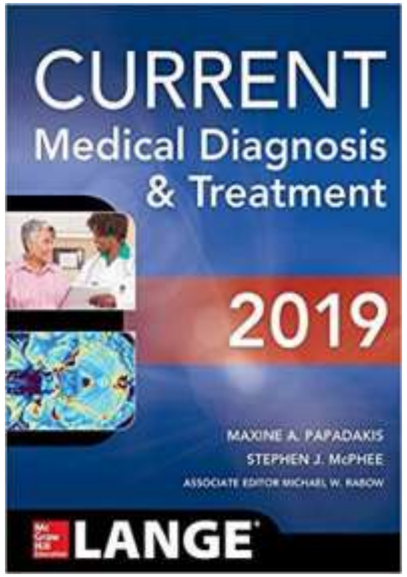
References & Further Readings

- **Kumar and Clark's Clinical Medicine,** 9th Edition
- **Davidson's Principles and Practice of Medicine**, 23rd Edition
- **Oxford Handbook of Clinical Medicine,** 10th Edition
- **CURRENT Medical Diagnosis and Treatment 2019 (LANGE CURRENT Series)** 58th Edition









Thank You!